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Uptake of perfluorooctanoate in freshly isolated hepatocytes from male and female rats

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ABSTRACT

Liver is a primary target organ for perfluorooctanoate (PFO, the deprotonated form of perfluorooctanoic acid, PFOA) distribution in both male and female rats. We studied the uptake of PFO in freshly isolated hepatocytes from male and female rats. We identified a non-saturable cell partitioning process for PFO using on-ice incubations. At 37 °C, hepatic uptake of PFO was composed of the non-saturable partition as well as a saturable, active uptake process. The K_m and V_{max} values for the active uptake process were 88.0 ± 9.1 μM and 5.61 ± 0.88 nmol/(min 10^6 cells), respectively, for male rat hepatocytes, and 76.1 ± 12.0 μM and 3.59 ± 0.29 nmol/(min 10^6 cells), respectively, for female rat hepatocytes. The values of PFO clearance by active uptake were 64.8 ± 15.7 and 47.6 ± 4.7 μL/(min 10^6 cells) for male and female rat hepatocytes, respectively. The active uptake of PFO in rat hepatocytes was inhibited by sulfobromophthalein, a known substrate of organic anion transporting polypeptides, with apparent inhibition constants of 85.9 ± 25.1 and 29.3 ± 19.2 μM in male and female rat hepatocytes, respectively. When serum albumin was added to the incubations, PFO hepatic uptake rates were reduced, but were proportional to the unbound fractions of PFO.

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1. Introduction

Perfluorooctanoic acid (PFOA) is a commercially important fluorochemical that has been used primarily in an ammonium salt form as a surface-active agent in the production of various fluoropolymers. At physiological conditions, PFOA is primarily in its deprotonated form as perfluorooctanoate (PFO). The mammalian toxicology and toxicokinetics of PFO have been studied extensively and were summarized in recent review papers (Kennedy et al., 2004; Lau et al., 2007; Andersen et al., 2008). PFO has been found in both wildlife and humans and is considered to be persistent in the environment (Prevedouros et al., 2006).

PFO half-life in human serum was estimated to be approximately 3.5 years (Olsen et al., 2007). In experimental animals, PFO was shown to be metabolically inert, distributed mainly to the liver and plasma, and excreted primarily via the urinary route (Vanden Heuvel et al., 1991; Kudo et al., 2001; Ohmori et al., 2003). The biological half-lives of PFO in experimental animals were considerably shorter than that in human, and showed significant interspecies variation (Hundley et al., 2006) and gender differences (Vanden Heuvel et al., 1991; Kudo et al., 2001, 2002; Ohmori et al., 2003; Hundley et al., 2006). PFO is considered to be actively excreted in the kidney, and the species and gender differences in PFO elimination have been attributed to the differences in renal transporters (Kudo et al., 2002; Andersen et al., 2006). Animal studies suggested that liver was the primary target organ for PFO toxicity (Kennedy et al., 2004), mainly via the activation of peroxisome proliferator-activated receptors (Vanden Heuvel et al., 2006). It was observed that, in the male rats, the overall proportion of PFO distributed to the liver (in relative to other tissues) decreased as hepatic concentration of PFOA increased within a dose range between 0.041 and 16.56 mg/kg body weight (Kudo et al., 2007). This observation strongly indicates a saturable, active transport mechanism for the hepatic uptake of PFO.

Rat liver active transport system is complex and composed of different classes of carrier proteins that are responsible for the uptake and efflux of organic anions with broad substrate specificity (Faber et al., 2003). The major classes of anion transporters in rat hepatocytes are organic anion transporting polypeptides (Oatps) (Faber et al., 2003). Preliminary results suggested that Oatp1a1 facilitated uptake of PFO in *Xenopus laevis* oocytes (Katakura et al., 2007), which implies the potential involvement of this transporter in hepatic uptake of PFO. Direct evidence of active hepatic uptake of PFO mediated by Oatps, however, is still lacking.

In this study, we investigated the hepatic uptake process for PFO using freshly isolated hepatocytes from male and female

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rats. For the first time, we were able to identify a high capacity, non-specific cell partitioning process for PFO at low-temperature (on-ice) incubation conditions, and showed that at 37 °C, hepatocyte uptake of PFO was composed of the non-specific partition as well as a saturable, active uptake process. We obtained and compared the kinetic parameters for the active uptake process for male and female rat hepatocytes. The active uptake of PFO in rat hepatocytes was inhibited by known substrates of Oatps and Oats. Concentration-dependent inhibition of PFO uptake by sulfobromophthalein (BSP) was further characterized. Since PFO primarily bound to serum albumin in rat plasma (Han et al., 2003), we also investigated PFO hepatic uptake in the presence of serum albumin, and showed that uptake of PFO in rat hepatocytes took place from the unbound fraction.

2. Experimental procedures

2.1. Materials

^{14}C-PFOA (>99%, labeled at the carboxyl carbon, ammonium salt; activity, 2.11 GBq/mmol) was purchased from Amersham Pharmacia Biotech (Piscataway, NJ). PFOA (>98%) was purchased from Oakwood Product (West Columbia, SC). Cell culture and perfusion media, buffers, and fetal bovine serum were obtained from Invitrogen (Carlsbad, CA). Fatty acid-free bovine serum albumin (BSA) and all other chemicals were from Sigma–Aldrich (St. Louis, MO).

2.2. Animals

Crl:CD(SD) rats were obtained from Charles River Laboratories (Raleigh, NC). Upon arrival, all animals were housed in quarantine for at least 4 days. Animals were provided tap water *ad libitum* and fed PMI® Nutrition International, LLC Certified Rodent LabDiet® 5002 *ad libitum*. At the time of hepatocyte isolation, rats were 6–8 weeks of age. Animal rooms were maintained at a temperature of 18–26 °C and a relative humidity of 30–70%, and were artificially illuminated (fluorescent light) on a 12-h light/dark cycle.

2.3. Hepatocytes isolation

The procedures for isolation of rat hepatocytes have been described previously (Mingoia et al., 2007). The isolated cells were suspended in Krebs–Henseleit buffer that was composed of 11.1 mM glucose, 1.18 mM $MgCl_2$ and KH_2PO_4, 4.7 mM KCl, 43.5 mM NaCl, 2.54 mM $CaCl_2$, and 25 mM $NaHCO_3$, pH 7.4. Cell viabilities were determined in the presence of 0.04% trypan blue and were greater than 85% in all cases.

2.4. Hepatocyte uptake assay

PFO hepatocyte uptake was measured according to a procedure described previously (Shitara et al., 2003) with slight modifications. Briefly, rat hepatocytes at 5×10^6 cells/mL in the Krebs–Henseleit buffer were pre-incubated at 37 °C for 10 min. At time zero, 200 μL of the cell suspension was added to an eppendorf tube containing 5 μL of ^{14}C-PFOA (416.25 Bq, final concentration = 1 μM). Incubation was carried out for up to 6 min at 37 °C. At 0.25, 0.5, 1, 2, 4, and 6 min, 100-μL aliquots of the cell suspension were placed in 0.5-mL eppendorf tubes that were previously layered with 200 μL of silicone/mineral oil mixture (9:1, v/v; d = 1.03 g/mL) and 50 μL of 2N NaOH. For substrate concentration-dependent experiments, the cells were incubated with PFOA ranging from 1 to 200 μM in the presence of 416.25 Bq ^{14}C-PFOA. 100-μL aliquots were taken at 30 s and were transferred to the top of the oil layer. The cells were immediately isolated by centrifugation at 14,000 rpm for 10 s. Samples were incubated at 50 °C overnight for complete lysis of the cells. After being frozen at −80 °C for 2 h, the tubes were cut through the oil layer. The cell pellet fraction was neutralized with 50 μL of 2N HCl, mixed with scintillation cocktail, and the radioactivity was determined in a liquid scintillation counter.

For inhibition experiments, the cells were incubated with 1 μM ^{14}C-PFOA in the presence of inhibitors (probenecid, salicylate, ketoglutarate, glutarate, azelate, taurocholate, and ketoprofen at 1 mM, and BSP at 250 μM). For determination of $K_{i,app}$ values for BSP, hepatocyte uptake of 1 μM ^{14}C-PFOA was measured in the presence of 5–250 μM BSP. The incubation was allowed for 2 min at 37 °C, and the cells were isolated with the oil-filtration technique that was described above.

Non-specific partitioning of PFO to rat hepatocytes was considered relatively temperature insensitive, comparing to the active uptake process. For determination of non-specific PFO partition to rat hepatocytes, the cells were suspended in a Krebs–Henseleit buffer that was refrigerated at 4 °C. The incubation was carried out on ice following the exact same procedure as outlined above for the 37 °C incubation.

For the uptake experiments in the presence of serum albumin, the cells were suspended in the Krebs–Henseleit buffer containing BSA. In the time-course experiments, 1 μM ^{14}C-PFOA was incubated at 37 °C with 5×10^6 cells/mL rat hepatocytes for up to 30 min at various BSA concentrations of 0, 1, 10, 100, and 600 μM. In the substrate concentration-dependent experiments, the cells were incubated with PFOA ranged from 25 to 500 μM in the presence of 5 μM ^{14}C-PFOA. The incubations were carried out at 37 °C for 2 min and at a BSA concentration of 600 μM.

2.5. Data analysis

Non-specific partitioning of PFO to rat hepatocytes followed a non-saturable, linear kinetics:

$$v_{par} = CL_{par}[S] \quad (1)$$

where v_{par} is the cell partition velocity of PFO (nmol/(min 10^6 cells)) that was determined by on-ice incubations, [S] is PFO concentration (μM), and CL_{par} is the non-saturable uptake clearance (μL/(min 10^6 cells)) and was obtained by linear regression to Eq. (1).

The total cellular uptake of PFO at 37 °C is considered to be the sum of the active uptake via organic anion transporters (saturable) and the non-specific partitioning (non-saturable):

$$v = \frac{V_{max}[S]}{K_m + [S]} + CL_{par}[S] \quad (2)$$

where v is the initial uptake velocity at 37 °C (nmol/(min 10^6 cells)), V_{max} is the maximum velocity (nmol/(min 10^6 cells)) for the active uptake, and K_m is the Michaelis constant (μM) for the active uptake. V_{max} and K_m were obtained by non-linear regression to Eq. (2) with the input of CL_{par} from Eq. (1).

When PFO concentration was much lower than the K_m, the inhibition of PFO uptake by BSP was described according to

$$v_i = \frac{v_0 - v_{par}}{1 + [I]/K_{i,app}} + v_{par} \quad (3)$$

where v_i and v_0 represent the initial PFO uptake velocities at 37 °C in the presence and absence of BSP, v_{par} was assumed to be independent of BSP concentration [I] (μM), and $K_{i,app}$ (μM), the apparent inhibition constant of BSP, was determined by non-linear regression to Eq. (3).

All curve fittings of non-linear regressions were done in software package Origin (version 7.0, OriginLab, MA).

3. Results

3.1. PFO uptake in rat hepatocytes

Uptake of 1 μM ^{14}C-PFOA in freshly isolated rat hepatocytes was time- and temperature-dependent (Fig. 1). Under on-ice incubation conditions, the uptake process was mainly driven by non-specific partition which reached equilibrium very quickly and the initial linear phase was not apparent after 15 s incubation. In contrast, PFO uptake rates increased proportionally to incubation time at 37 °C. The initial linear phase for the uptake rates at 37 °C was main-

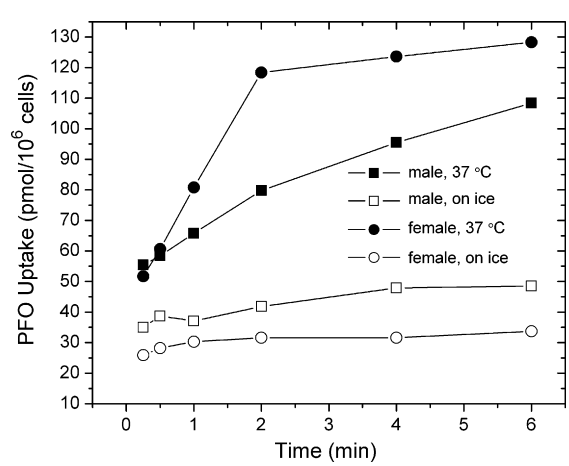

Fig. 1. PFO uptake by male (squares) and female (circles) rat hepatocytes as a function of time. 1 μM of ^{14}C-PFOA was incubated with 5×10^6 cells/mL hepatocytes on ice (open symbols) and at 37 °C (filled symbols). The data were averaged from triplicate incubations. The coefficient of variation values were below 10%.

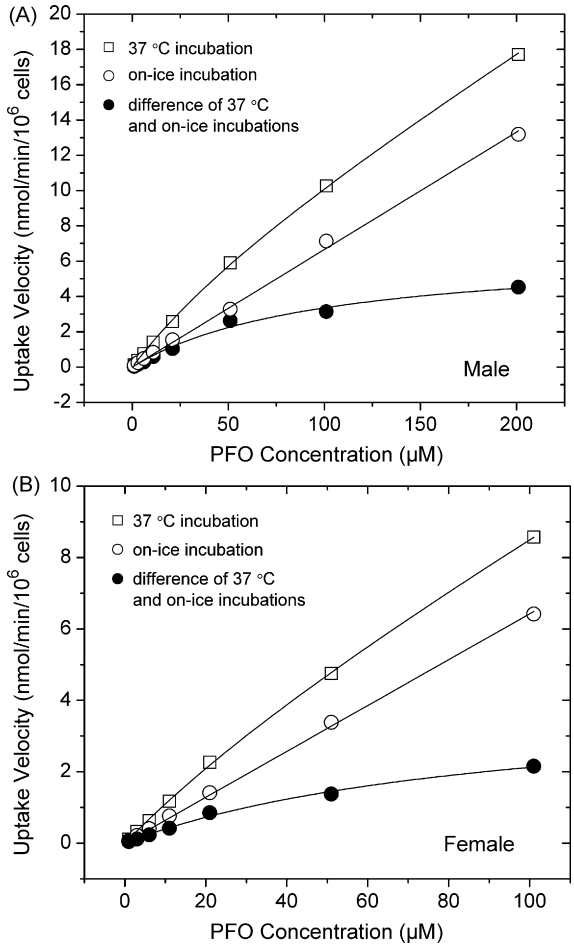

Fig. 2. Representative velocity *vs.* PFOA concentration curves for the uptake of PFO in male (A) and female (B) rat hepatocytes at 5×10^6 cells/mL. The incubations were carried out for 30 s on-ice (open circles, curve-fit to Eq. (1)) and at 37 °C (open squares, curve-fit to Eq. (2)). Their calculated difference was plotted with filled circles and fitted to the Michaelis–Menten equation. Kinetic parameters obtained from these curves are reported in Table 1 and were calculated from three individual hepatocyte isolations with pooled cells from at least two animals per isolation.

tained for at least 2 min of incubation for both male and female rat hepatocytes.

The kinetics of PFO uptake in rat hepatocytes were studied at both on-ice and 37 °C incubation conditions for PFO concentrations ranging from 1 to 200 μM. The uptake velocities exhibited a linear relationship with PFO concentration in both male (Fig. 2A) and female (Fig. 2B) rat hepatocytes when incubated on ice. This linear relationship is explained by non-specific partitioning of PFO to the cells and was described by Eq. (1) in Section 2. The values of PFO uptake clearance due to cell partitioning (CL_{par}) were obtained from 30 s incubation on ice and were 67.8 ± 6.4 and 60.8 ± 3.2 μL/(min 10^6 cells) for male and female rat hepatocytes, respectively (Table 1). Their difference was not statistically significant according to the calculated *P*-value (Table 1).

PFO uptake by rat hepatocytes was faster at 37 °C than on-ice incubations, but did not reach saturation even at the highest PFO concentration (200 μM, Fig. 2). The total uptake of PFO at 37 °C was considered to be the sum of an active uptake process and the non-specific partitioning that was demonstrated from on-ice incubations. Eq. (2) was used to describe the uptake kinetics at 37 °C and the obtained kinetic parameters were listed in Table 1. The K_m and V_{max} values for the active uptake process were determined to be 88.0 ± 9.1 μM and 5.61 ± 0.88 nmol/(min 10^6 cells), respectively, for male rat hepatocytes, and 76.1 ± 12.0 μM and 3.59 ± 0.29 nmol/(min 10^6 cells), respectively, for female rat hepatocytes. The K_m values were not statistically different between male and female, but the male V_{max} value was significantly larger than the female's ($P < 0.05$, Table 1). PFO clearance by active uptake (defined by V_{max}/K_m) was slightly higher in male rat hepatocytes (64.8 ± 15.7 μL/(min 10^6 cells)) than that in the females (47.6 ± 4.7 μL/(min 10^6 cells)), although the difference was not statistically significant ($P > 0.05$, Table 1).

The difference curves (velocities from 37 °C incubations minus velocities from on-ice incubations) were also plotted in Fig. 2 and were adequately described by a saturable process with the Michaelis–Menten equation. The kinetic parameters (K_m and V_{max}) determined from the difference curves (data not shown) were consistent with the values obtained from Eq. (2).

3.2. Inhibition of PFO uptake in rat hepatocytes

In order to evaluate the specificity of PFO uptake in rat hepatocytes, we performed an inhibition study using known substrates for Oatps and Oats. The uptake of 1 μM ^{14}C-PFOA in rat hepatocytes at 37 °C was determined in the presence or absence of 1 mM probenecid, salicylate, ketoglutarate, glutarate, azelate, taurocholate, ketoprofen, and 250 μM BSP. As shown in Fig. 3, BSP exhibited the most inhibition activities for both male and female rat hepatocytes and almost completely abolished PFO uptake at 250 μM. Inhibitions from other compounds at 1 mM were much less significant. Glutarate, ketoglutarate and salicylate only showed negligible inhibition activities.

3.3. Concentration-dependent inhibition of PFO uptake in rat hepatocytes by BSP

Inhibition of PFO uptake by BSP was concentration-dependent. Fig. 4 shows the percentage of active uptake of 1 μM ^{14}C-PFOA as a function of BSP concentrations from 5 to 250 μM. The $K_{i,app}$ value determined in male rat hepatocytes (85.9 ± 25.1 μM) was significantly higher than that in female rat hepatocytes (29.3 ± 19.2 μM, $P < 0.05$).

3.4. PFO uptake in rat hepatocytes in the presence of serum albumin

In blood, greater than 90% of PFO bound to serum albumin with a dissociation constant (K_d) value approximately 0.3 mM (Han et al., 2003). In order to evaluate albumin binding effect on hepatic

Table 1
Kinetic parameters[a] for PFO uptake in freshly isolated rat hepatocytes

	CL_{par} (μL/(min 10^6 cells))	K_m (μM)	V_{max} (nmol/(min 10^6 cells))	V_{max}/K_m (μL/(min 10^6 cells))
Male	67.8 ± 6.4	88.0 ± 9.1	5.61 ± 0.88	64.8 ± 15.7
Female	60.8 ± 3.2	76.1 ± 12.0	3.59 ± 0.29	47.6 ± 4.7
P-value[b]	0.17	0.24	0.02	0.14

[a] Values were obtained from three individual hepatocyte isolations with pooled cells from at least two animals per isolation.
[b] For the comparison between male and female; obtained from independent Student's *t*-tests.

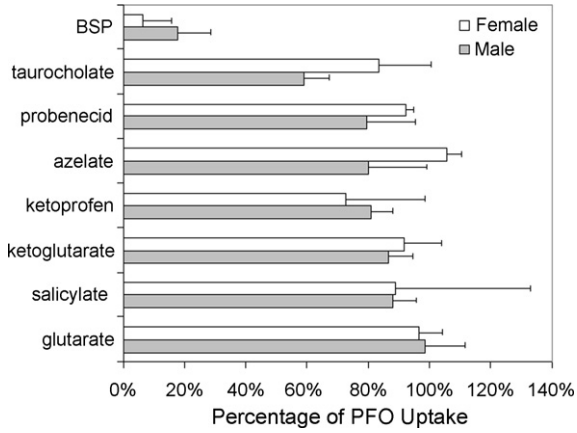

Fig. 3. Inhibition of 1 μM ^{14}C-PFOA uptake in rat hepatocytes at 37 °C by various known substrates of anion transporters. The concentrations of the inhibitors were 1 mM for probenecid, salicylate, ketoglutarate, glutarate, azelate, taurocholate, and ketoprofen, and 250 μM for BSP. The values were presented as the percentage of PFO uptake in the absence of the inhibitors. Uptake rate contributed from the non-specific partition was obtained from on-ice incubations and considered non-inhibitable, therefore, was subtracted out from the total uptake rate before the percentage calculation. Error bars represent standard deviations from three individual experiments.

uptake of PFO, we obtained uptake rates of 1 μM ^{14}C-PFOA in rat hepatocytes as a function of time in the presence of BSA at different concentrations. Fig. 5 shows that for both male and female rat hepatocytes, PFO uptake rates were reduced with the increase of BSA concentrations. We also estimated the free, unbound concentrations of PFO in the presence of 10, 100, and 600 μM BSA assuming a K_d value of 0.3 mM and seven identical PFO binding sites on BSA (Han et al., 2003). PFO uptake rates appeared to have a linear relationship with the unbound PFO concentrations (insets, Fig. 5).

In the presence of physiological concentration of albumin (600 μM), we studied PFO concentration-dependent uptake kinetics at both on-ice and 37 °C incubation conditions (Fig. 6). The on-ice incubation curves significantly deviated from linear relationship with the estimated free PFO concentration at PFO doses

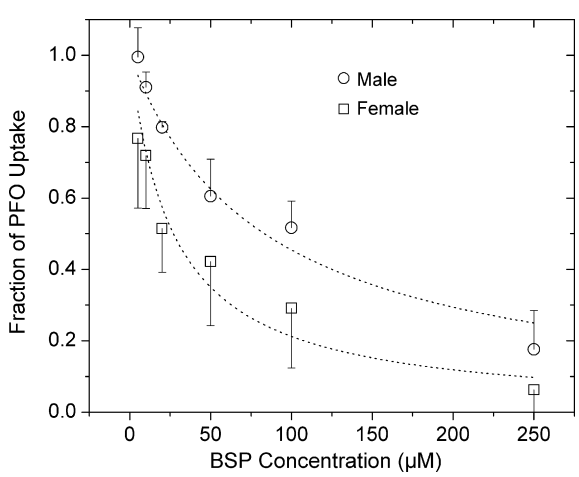

Fig. 4. Percentage of active uptake of 1 μM ^{14}C-PFOA in male (circle) and female (square) rat hepatocytes as a function of BSP concentration. The incubations were carried out for 2 min at 37 °C and at a cell concentration of 5×10^6 cells/mL. The non-inhibitable component (v_{par}) was determined from on-ice incubations. The dotted lines showed the non-linear regressions of Eq. (3). The data were obtained from three individual hepatocyte isolations with pooled cells from at least two animals per isolation.

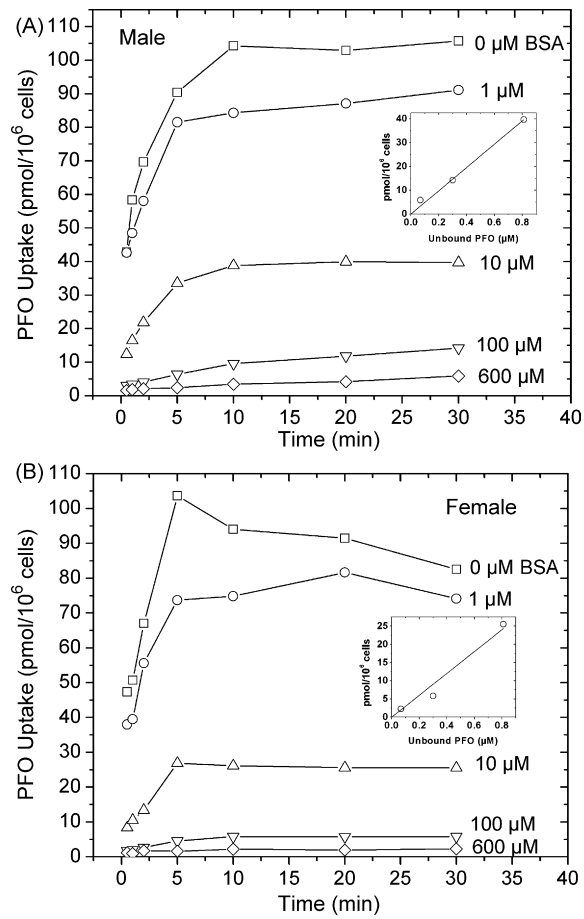

Fig. 5. Time-course PFO uptake by male (A) and female (B) rat hepatocytes in the presence of 0, 1, 10, 100, and 600 μM BSA. 1 μM of ^{14}C-PFOA was incubated with 5×10^6 cells/mL hepatocytes at 37 °C for up to 30 min. The insets show the relationship between the uptake rates and the calculated free concentrations of PFO at BSA concentrations of 10, 100, and 600 μM. Free concentrations of PFO were calculated based on 0.3 mM K_d and assumption of seven identical PFO binding sites on BSA (Han et al., 2003). Data were the average of duplicate samples.

greater than 500 μM (corresponding to 37 μM unbound PFO based on the assumptions of binding parameters discussed above, data not shown). Therefore, we have limited coverage of free PFO concentration range that was below the K_m (Fig. 6). Nevertheless, we were able to show that for both male and female rat hepatocytes, active PFO uptake rates (difference between 37 °C and on-ice incubations) had linear correlation to the concentrations of unbound PFO (filled circles, Fig. 6). Our evidence supports a mechanism of passive binding of PFO to albumin and only the free, unbound fraction of PFO is available for the uptake in rat hepatocytes.

4. Discussion

PFO is a metabolically inert compound that exhibits sex-, species-, and dose-related differences in its toxicokinetic properties. As suggested in previous studies (Kudo et al., 2002; Andersen et al., 2006; Ljubojevic et al., 2007), the mechanism of PFO transport at the cellular level might hold the key to our understanding of the distribution, disposition, and elimination of PFO. In this study, for the first time, we showed evidence of hepatic uptake of PFO by active transporters. In addition, we revealed a passive diffusion mechanism for PFO uptake via a non-specific partition process.

The non-specific partition of PFO to rat hepatocytes was identified using an on-ice incubation procedure (Fig. 2) and was

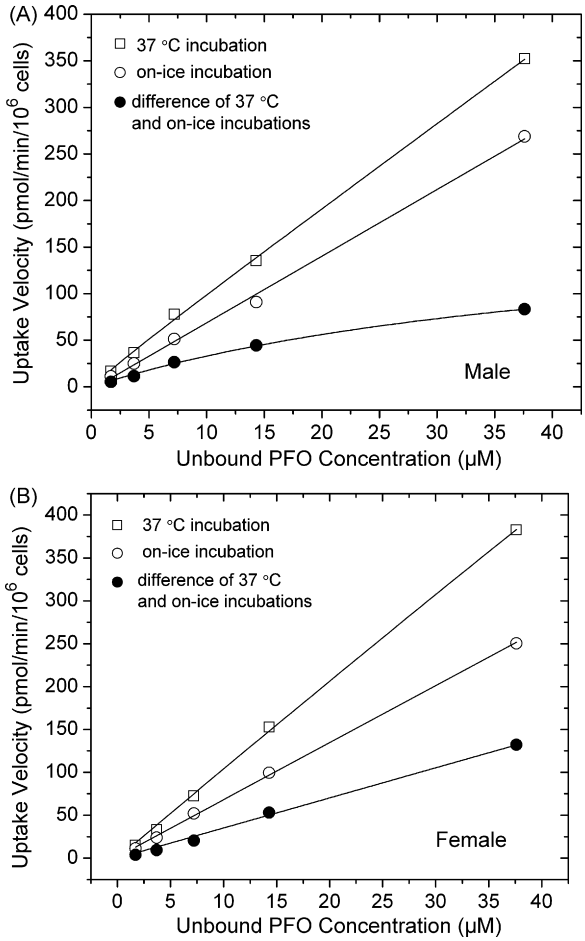

Fig. 6. Velocity *vs*. PFO concentration curves for the uptake of PFO in male (A) and female (B) rat hepatocytes at 5×10^6 cells/mL and in the presence of 600 μM BSA. The incubations were carried out for 2 min on-ice (open circles) and at 37 °C (open squares). Their calculated difference was plotted with filled circles. Data were the average of duplicate samples.

adequately described by a linear function of Eq. (1). A similar non-saturable, passive diffusion process has been described previously for long-chain fatty acids (Abumrad et al., 1998), BSP (Glavy et al., 2000), cerivastatin (Shitara et al., 2003), temocaprilat (Hasegawa et al., 2003), salicylate (Hasegawa et al., 2003), and imipramine (Hallifax and Houston, 2006) using the same linear relationship (Eq. (1)), or for BSP (Marchegiano et al., 1992) and resveratrol (Lancon et al., 2004) using 4 °C incubations. It is worth noting that the passive diffusion process determined in our assay cannot be distinguished between the actual transport of PFO across the plasma membrane and another possibility of simple association/binding of PFO to the membrane. The contribution of this passive diffusion component to the total amount of PFO uptake was significant, accounting for approximately 50–60% of the total uptake clearance (Table 1) at relatively low PFO concentrations ($<K_m$), and became even more important at higher PFO concentrations that saturated the active uptake process (Fig. 2). PFO hepatocyte-buffer partition coefficient ($K_{hb} = A_{cell}/A_{buffer}$, where A_{cell} and A_{buffer} are the amount of radioactivity associated with the cell fraction and the supernatant fraction at equilibrium, respectively) was estimated with on-ice incubations to be approximately 0.3 for both male and female rat hepatocytes at a cell concentration of 5×10^6 cells/mL. PFO dipalmitoylphosphatidylcholine (DPPC)–water partition coefficient ($K_{lw} = A_{lipid}/A_{water}$, where A_{lipid} and A_{water} are the mole amount of PFO associated with the DPPC lipid bilayer phase and the water phase at equilibrium, respectively) was measured indirectly based on the phase transition characteristics of the DPPC bilayers in the presence of PFO and fluorescence probes, and the value was determined to be $(1.5–2.1) \times 10^4$ (Xie et al., 2007). The large difference between K_{hb} and K_{lw} is mainly due to the fact that both K_{hb} and K_{lw} were ratios of the amount of PFO in cell (or lipid) phase *vs*. in buffer (or water) phase, which, in contrast to concentration ratios, are not a constant and dependent on cell (or lipid) concentrations in the aqueous solution. Therefore, direct comparison of these two values is not possible. Furthermore, K_{lw} was measured in pure water (Xie et al., 2007). This unbuffered condition may cause large shift on the pH value of the solution. pH value determines the ionization state of PFO ($pK_a = 2.8$, Brace, 1962) which in turn affect greatly the lipophilicity and water solubility of PFO (or PFOA). In contrast, our work was done in an adequate buffered condition to ensure the final pH of approximately 7.4.

Kudo et al. (2007) have revealed that PFO distribution in male rat liver is dose-dependent. The percentage of PFO distributed to the liver, in relative to other tissues, was determined after 2 h of i.v. injection of 6 dose levels ranging from 0.041 to 16.56 mg/kg body weight. The percentage value decreased from 52% at the dose of 0.041 mg/kg body weight to 27% at the dose of 16.56 mg/kg body weight. The measured PFO liver concentrations at these two dose levels were 0.558 and 87.45 μg/g liver, respectively (Table 1 in Kudo et al., 2007), which are equivalent to 1.35 and 211 μM of PFO concentrations (by assuming liver density of 1 g/mL). Analogously, PFO blood concentrations at these two dose levels (0.041 and 16.56 mg/kg) were 0.319 and 139 μM, respectively (Table 1 in Kudo et al., 2007). In this study, we determined that the K_m value of PFO active uptake in male rat hepatocytes was 88 μM (Table 1). It becomes apparent that, at the low dose 0.041 mg/kg in Kudo et al.'s study (Kudo et al., 2007), the PFO concentrations in blood and in the liver were both significantly lower than the K_m value, and therefore, active uptake should have played an important role in the total PFO hepatic uptake. It appears that at the high dose 16.56 mg/kg, the PFO concentrations in blood and liver were greater than the K_m. However, since PFO is mainly albumin-bound in plasma, the free, unbound PFO concentration is calculated approximately 10 μM at the 16.56 mg/kg dose level (Han et al., 2003), which is still significantly lower than the K_m value for the active hepatic uptake. Therefore, the non-linear, dose-dependent PFO liver distribution observed by Kudo et al. cannot be completely explained by our study.

We observed that PFO uptake velocities in the presence of 600 μM BSA (Fig. 6) were significantly lower than those measured in the buffer (Fig. 2) at the same concentration range of unbound PFO (actual dose in Fig. 2 and calculated in Fig. 6). The exact reason for this observation is unclear, but we assume at least partial contribution to this difference from the estimation of the free PFO concentration in the presence of BSA. The free PFO concentration in the presence of BSA (Fig. 6) was calculated based on human and rat albumin-PFO binding data (Han et al., 2003). Species difference in PFO-albumin binding is possible and our estimation of the free PFO concentration may be subject to significant error for this reason. Nevertheless, Fig. 5 unarguably demonstrated that binding to BSA indeed significantly reduced PFO hepatic uptake velocity.

The active uptake process for PFO had comparable K_m values for male and female rat hepatocytes (Table 1). The V_{max} value, however, was significantly higher in male rat hepatocytes than that in the females (approximately 1.6-fold, Table 1), which correlates well with the relationship between gender and expression levels of Oatp1a4 (old name Oatp2, Rost et al., 2005) and Oat3 (Buist et al., 2002; Kobayashi et al., 2002) in rat liver. Oatp1a4, however, is not a promising candidate for PFO transport, because as shown in Fig. 3, PFO hepatic uptake was significantly inhibited by BSP which,

reportedly, is not a substrate of rat Oatp1a4 (Reichel et al., 1999). Rat Oat3 has just recently been shown to facilitate the transport of PFO in *X. laevis* oocytes (Katakura et al., 2007), and in Oat3-transfected HEK293 cells (Nakagawa et al., 2008). The apparent K_m value for PFO uptake in Oat3-transfected HEK293 cells was reported to be 80.2 ± 17.8 μM (Nakagawa et al., 2008), which is in good agreement with the K_m values we determined for rat hepatocytes (Table 1). Therefore, Oat3 may, at least to some extent, have contributed PFO active uptake in rat liver. Additionally, Fig. 3 shows that, among eight known substrates/inhibitors of Oats and Oatps, BSP was most potent in PFO uptake inhibition. It has been reported that BSP is a substrate of Oatp1a1 (old name Oatp1, Jacquemin et al., 1994; Meier et al., 1997; Hata et al., 2003) and, furthermore, PFO uptake was significantly enhanced by Oatp1a1-induced *X. laevis* oocytes (Katakura et al., 2007). Therefore, Oatp1a1 is another promising candidate for PFO transport in rat liver.

In summary, we presented experimental evidence of rat hepatic uptake of PFO via a carrier protein-mediated active uptake mechanism. We also showed that the process of passive diffusion contributed significantly to the overall hepatic uptake of PFO. Albumin-binding is likely a passive process to limit the free fraction of PFO that is available for hepatic uptake. A known organic anion transporter substrate, BSP, effectively inhibited the active hepatic uptake of PFO. BSP, therefore, could become a useful tool for future elucidation of the mechanism involved in PFO membrane transport process in both liver and kidney.

Conflict of interest

The authors declare that there are no conflict of interests.

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